CDE4301 Innovation & Design Capstone



Interim Report

Project Code: ASI-402

Title: Design of a Custom Quadcopter Subsystem for the NUS Mars Rover Team - University Rover Challenge 2027

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Acknowledgements

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Executive Summary

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List of Abbreviations

Abbreviation	Meaning

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1. Introduction

The University Rover Challenge (URC) is a robotics competition for university students, organized annually by the Mars Society near the Mars Desert Research Station (MDRS) in the desert of southern Utah, United States. The competition tasks student teams with designing and building next-generation Mars exploration rovers capable of executing field missions that simulate real planetary surface operations.

Field performance is evaluated across 4 distinct missions—Delivery, Autonomy, Equipment Servicing, and Astrobiology—each worth 100 points for a maximum achievable score of 400 mission points, though the drone can only be utilized for the Delivery and Autonomy missions. The Delivery Mission requires rovers to retrieve and deliver objects weighing up to 5 kg across challenging terrain over distances up to 1 km. The Autonomy Mission demands full autonomous navigation through waypoints as well as astronaut assistance and tool retrieval tasks without any human teleoperation. Overall team standings combine field mission points with engineering evaluations from the System Acceptance Review (SAR), which is submitted prior to the field competition.

1.1 Project Overview & Problem Statement

For this competition, our rover is expected to traverse undulating terrain at an estimated distance of up to 2 km. This introduces the risk of communication loss between the rover and our base station due to signal attenuation over long ranges and obstructed line-of-sight due to natural obstacles. As such, the aerial drone subsystem is deployed specifically during the Delivery Mission and Autonomy Mission to support ground operations. Serving as a Non-Line-of-Sight (NVLOS) RF relay operating at an elevated hover altitude of 60 m AGL, the drone acts as an airborne repeater bridging primary telemetry and video links between the Command and Control base station and the ground rover when line-of-sight is severed by canyon walls and terrain ridges. Additionally, the drone provides aerial reconnaissance by scouting ahead of the ground rover to identify optimal driving paths, locate scattered mission equipment, and detect elevated ArUco markers beyond the field of view of ground-level cameras.

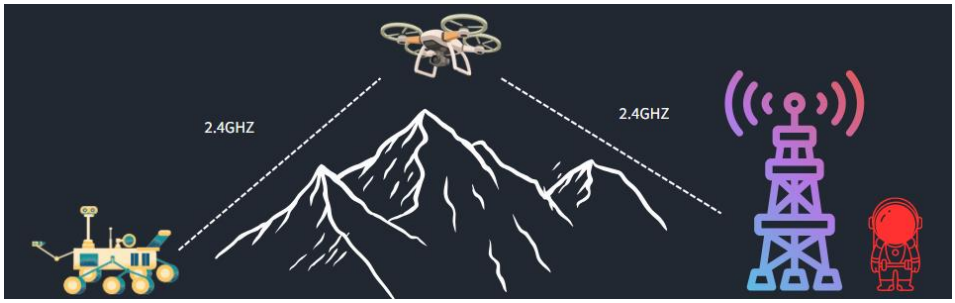


Figure 1: Elevated drone communication relay bypassing terrain obstructions to maintain an uninterrupted 2.4 GHz link between the base station and rover.

A flat quadcopter layout with fixed-pitch propellers was selected over complex coaxial or hexacopter platforms. This design minimizes dry structural weight, eliminates mechanical single-point failure modes in fine desert dust, and delivers high hover aerodynamic efficiency under strict competition mass rules.

1.2 URC2027 Competition Missions & Restrictions

To simulate flight performance under lower atmospheric density, the drone must always carry a non-functional dummy mass equal to its battery weight. The total weight is capped at 5.0 kg, which includes both the dummy payload and a FAA Remote ID module around 20g. Operating altitude is restricted to a maximum of 120 m above ground level in compliance with FAA regulations.

Environmental conditions at the Mars Desert Research Station feature summer temperatures reaching up to 46 degrees Celsius based on historical climate data, placing severe thermal management demands on onboard electronics. Operational wind guidelines strongly recommend that the platform withstand sustained desert winds of 48 km/h, requiring a minimum top speed of 36 km/h when carrying the dummy payload. The entire competition spans a 1-mile square near the Mars Desert Research Centre.

1.3 Mars Rover Team

Below is the breakdown of the entire Mars rover team and its subsystem.
(insert image)

1.4 Team Workload Breakdown

Subsystem responsibilities are divided among four team members based on engineering specialization. Julian serves as Team Lead, overseeing communications equipment selection and Non-Line-of-Sight RF relay architecture. Joel manages electrical hardware and power distribution. Benedict handles propulsion system sizing, motor-propeller thrust matching calculations, and hover aerodynamic efficiency optimization. Zi Hann is responsible for mechanical structures and structural carbon fiber layout.

1.5 Overall Design Objective

The principal objective of the drone subsystem is to deliver a robust aerial asset tailored specifically to URC mission constraints. The platform is engineered to achieve a sustained hover endurance of at least 20 minutes at 5.0 kg MTOW while carrying the 1:1 dummy mass. It targets an operational visual reconnaissance radius of 2 km and maintains uninterrupted RF signal relay coverage over a 1 km operating radius. By carrying an airborne repeater module, the drone bridges telemetry and video feeds between the ground rover and base station while transmitting HD video streams for real-time terrain mapping and elevated ArUco marker identification.

2. Literature Review

The team's predecessor platform suffered from structural, electrical, and scaling bottlenecks. The baseline platform utilized a 580 mm wheelbase frame equipped with standard 4008 motors, 15-inch folding propellers, and a 6S 10,000 mAh Li-Ion pack flying at a total weight of 2,654 g. Combined four-motor thrust testing revealed a 27.1% thrust deficit between isolated bench tests (1,900 g per motor) and full system integration (1,385 g per motor). Analytical decomposition confirmed that battery voltage sag under a 39.7A load accounted for 20.8% of this reduction, while turbulent inter-motor propeller wash interference on the compact frame accounted for the remaining 7.9%.

The redesigned platform addresses these empirical loss mechanisms by optimizing mass distribution, expanding frame geometry to an 800 mm carbon fiber wheelbase to eliminate rotor wash overlap, and utilizing short 10 AWG silicone power leads with anti-spark XT90-S interconnects to minimize internal resistance losses. Avionics integration was upgraded to Bidirectional DShot telemetry with dynamic PID notch filtering, suppressing mechanical frame resonance and reducing motor waste heat by around 15 degrees Celsius in ambient desert conditions.

3. Technical Design

3.1 Propulsion System

3.1.1 Design Requirements and Methodology

The quadcopter is designed for communication relay and terrain surveying, with a maximum take-off mass (MTOW) of 5 kg. Based on these requirements, preliminary propulsion sizing was conducted to estimate the required motor thrust and propeller diameter, followed by the selection and comparison of motor–propeller combinations alongside the airframe design. A mass budget was then developed using the estimated component masses to determine the allowable battery mass.

The design follows an iterative approach, integrating propulsion sizing, mass budgeting, battery selection, and mechanical integration. Changes in component or battery mass affect the total aircraft mass, required thrust, power consumption, and predicted endurance. Finally, a battery mass sensitivity analysis and endurance optimisation were performed to estimate flight time and select a suitable configuration while satisfying the 5 kg MTOW constraint.

3.1.2 Preliminary Propulsion Sizing

Hover and Maximum Thrust Requirement

At the maximum take-off mass (MTOW) of 5 kg, each of the four motors must provide 5,000/4 = 1,250 gf of hover thrust (approximately 12.26 N). A preliminary design thrust-to-weight ratio of 2.0 gives a target maximum available thrust of at least 2,500 gf per motor. This reserve supports climb and manoeuvring capability, although actual wind tolerance also depends on control authority and battery-limited thrust.

Disk Loading and Propeller Diameter

Disk loading is defined as the aircraft weight divided by the total rotor disk area: $DL = \frac{W}{A_{total}}$, where W = mg. For N equal rotors, $A_{total} = N \cdot \pi \cdot \frac{D^2}{4}$ and $D = \sqrt{\frac{4mg}{N \cdot \pi \cdot DL}}$. A relatively low preliminary disk loading of 75 N/m² was adopted for this endurance-oriented quadcopter. Lower disk loading generally reduces ideal induced hover power, but requires larger propellers and consequently a larger airframe, which may increase structural mass and expose more area to wind. Conversely, higher disk loading permits a more compact airframe but increases ideal induced power. Therefore, 75 N/m² was selected as a practical compromise between hover efficiency and airframe size. At 5 kg, the required total disk area is approximately 0.654 m², corresponding to an 18-inch propeller diameter.

Disk loading	Diameter at 5 kg	Design implication
50 N/m2	22.0 in	Lower ideal induced power, but larger propellers/frame; potentially greater wind-exposed area
75 N/m2	18.0 in	Balanced preliminary choice for hover efficiency and practical frame size
100 N/m2	15.6 in	Smaller frame, but higher ideal induced power; less favourable for endurance

The selected disk loading serves only as an initial sizing guideline. The actual motor–propeller combination was subsequently evaluated using manufacturer performance test data, including hover power, thrust efficiency and maximum tested thrust.

3.1.3 Motor and Propeller Selection

Comparison of Shortlisted Motor–Propeller Combinations

Based on the preliminary propulsion sizing, multiple motor–propeller combinations were shortlisted, with the two most promising candidates presented for further comparison: the T-Motor MN501-S KV360 paired with the P18×6.1 propeller and the MAD 5008 EEE 340KV paired with the FLUXER PRO 18×6.1 propeller. Both combinations were evaluated using manufacturer performance data. The electrical input power at the required hover thrust of 1,250 gf per motor was estimated using linear interpolation between adjacent test points.

Parameter	T-Motor MN501-S	MAD 5008 EEE 340KV
Motor mass	175 g	127 g
Power at 1,250 gf	139.0 W	124.9 W
Thrust efficiency	9.0 gf/W	10.0 gf/W
Maximum tested thrust	4,644 gf	3,797 gf

At the required hover thrust, the MAD motor consumes approximately 10.2% less input power and delivers 11.3% higher thrust efficiency. Its 48 g lower motor mass results in a total mass saving of approximately 192 g across four motors (27.4%). Although its maximum tested thrust is lower, it can still generate 3,797 gf per motor, exceeding the preliminary requirement of 2,500 gf. Based on its lower power consumption, higher thrust efficiency, and reduced mass, the MAD Motor 5008 EEE KV360 was selected. However, these comparisons are based on different manufacturer test setups and should therefore be considered preliminary

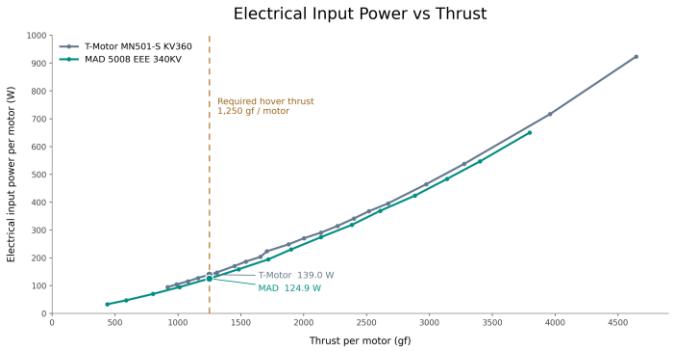


Figure x. Manufacturer power-thrust data and the 1,250 gf hover reference.

Comparison of Different Propeller Sizes

The manufacturer provides performance data for three FLUXER PRO propellers paired with the MAD 5008 EEE 340KV motor: 17×5.8, 18×6.1, and 19×5.7, all tested under a 6S configuration. These combinations were compared at the required hover thrust of 1,250 gf per motor using linear interpolation between adjacent test points.

The 17-inch propeller has nearly identical hover power consumption to the 18-inch propeller but produces lower maximum thrust (3,365 gf versus 3,797 gf) and requires a higher hover throttle setting (54% versus 50%). Given the uncertainty in wind conditions at the survey site, the **1**8×6.1 propeller was preferred for its greater tested thrust reserve, while maintaining comparable hover efficiency.

Propeller	Power at 1,250 gf	Maximum tested thrust
17x5.8	~126.2 W (~54% throttle)	3,365 gf
18x6.1 (selected)	~124.9 W (~49.8% throttle)	3,797 gf
19x5.7	~121.2 W (~48.1% throttle)	3,924 gf

The 19-inch propeller offers slightly lower hover power consumption than the 18-inch option but requires greater rotor clearance, potentially resulting in a larger and heavier frame. However, its maximum tested thrust increases by only 3.3% (from 3,797 gf to 3,924 gf), offering a modest improvement.

Therefore, the 18×6.1 propeller was selected as a practical compromise between hover efficiency, thrust reserve, and frame integration. This selection also agrees with the 18-inch diameter estimated using the preliminary disk loading of 75 N/m², providing a useful cross-check of the initial sizing. The 6.1-inch pitch was adopted from the manufacturer-tested combination.

Additionally, the calculated pitch-to-diameter ratio (P/D) of 0.339 indicates a relatively low pitch, consistent with the endurance-focused mission requirements.

3.1.4 Battery Selection and Endurance Optimisation

Battery Mass Sensitivity Analysis

Following the preliminary propulsion-system selection, a battery mass sensitivity analysis was conducted to investigate the competing effects of battery mass on available energy and required propulsion power. Under the competition requirement, the dummy payload mass must equal the battery mass. Although a heavier battery provides greater stored energy, it also increases the hover thrust, and electrical power required to support the aircraft.

To study this trade off, a parametric sensitivity analysis was performed across candidate battery masses. For each case, the total aircraft mass (including equal dummy mass), required thrust, motor operating point, electrical power demand and predicted hover endurance were recalculated using manufacturer performance data. A constant specific energy of 287 Wh/kg was assumed, representative of the selected high-energy lithium-ion battery class (355.2 Wh / 1.24 kg = 286.5 Wh/kg). This enables a consistent comparison of the mass–energy trade-off without introducing differences between individual battery products.

As illustrated in Figure 2, predicted hover endurance increases across the feasible battery mass range, although the incremental benefit decreases as battery mass rises. This diminishing-return behaviour occurs because the gain in usable battery energy is increasingly offset by the additional propulsion power needed to lift the battery and dummy payload. Therefore, battery capacity was maximised within the 5 kg maximum take-off mass (MTOW) constraint. The selected battery mass is determined by this constraint and the availability of suitable commercial batteries, rather than by the unconstrained maximum of the endurance curve.

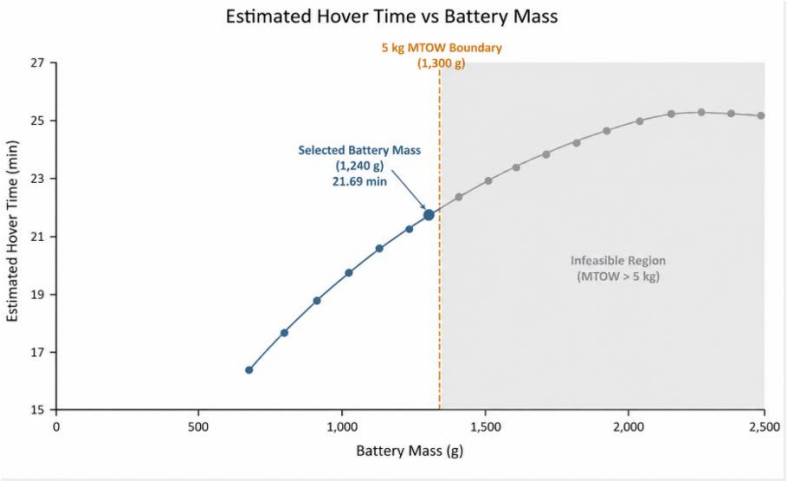


Figure 2. Predicted hover endurance versus battery mass. The grey region denotes configurations exceeding the approximate 5 kg MTOW boundary.

3.1.5 Selected Battery and Technology Rationale

Based on the battery mass sensitivity analysis and the propulsion system's 6S configuration, the MAD 6S 16 Ah solid-state lithium-ion battery was selected. Its specifications are summarized in the table below.

Battery specification	Selected value
Battery model	MAD 6S 16 Ah solid-state lithium-ion
Configuration / nominal voltage	6S / 22.2 V
Capacity / nominal energy	16 Ah / 355.2 Wh
Confirmed battery mass	1,240 g
Specific energy (nominal)	~286.5 Wh/kg
Continuous discharge	3C = 48 A
Maximum discharge	5C = 80 A

A high-energy lithium-ion battery was preferred over a conventional high-discharge lithium-polymer (LiPo) battery due to its higher specific energy, allowing greater energy storage for a given mass. Although LiPo batteries generally offer higher discharge capability, lithium-ion batteries are more suitable for our endurance-focused hover and terrain-survey missions, provided that the required discharge current can be supported.

3.1.6 Endurance Model Assumptions and Validation Plan

To provide a conservative preliminary endurance estimate, several assumptions were adopted, as summarised in the following table. These assumptions provide the rover team with an initial flight-time estimate to support mission planning and task allocation.

Assumption	Value / treatment
Battery specific energy for sensitivity sweep	287 Wh/kg (constant)
Dummy payload	Equal to battery mass
ESC efficiency	90% (assumed)
Wiring/electrical-path efficiency	90% (assumed)
Battery energy reserve	15% (assumed)
Auxiliary electrical load	60 W
Mass growth allowance	5% (design case; avoid duplicate application)

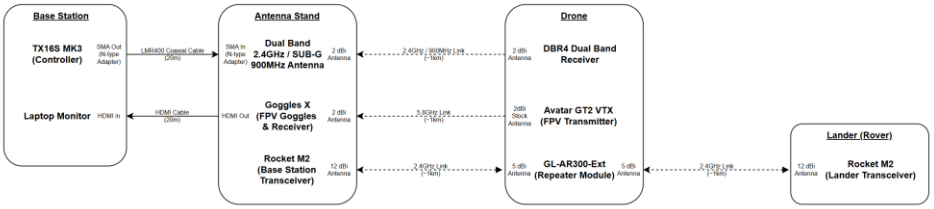
Once the motor, propeller, and battery are available for testing, thrust and electrical power consumption will be measured at representative operating points to establish an experimental performance table. The results will then be used to develop a more realistic mission-based endurance model covering take-off, climb, transit, terrain surveying, hover and descent, providing an improved flight-time prediction for mission planning.

3.2 Communication System Selections

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3.2.1 Signal Architecture

The communication links are our vital architecture for the competition. This section details the design of the non-direct line-of-sight communications link between the rover, base station and drone, as well as the architecture for ensuring live video feed and radio control with the drone.



We split the communication systems into three independent wireless links, each serving a distinct operational function. This allows us to design and evaluate the control, video and rover telemetry separately within their own frequency band, accounting for latency and reliability.

Link	Frequency	Selected Components	Function
Remote Control (RC)	2.4GHz	Radiomaster TX16S MK3 Controller + DBR4 Receiver	Transmit flight control commands and receiver telemetry.
First Person View (FPV) Video	5.8GHz	Walksnail Avatar GT2 Camera Kit + Goggles X	Real-time video transmission for reconnaissance and Situational awareness
Rover Telemetry	2.4GHz	Rocket M2 + GL-AR300M-Ext Repeater	Proposed communication relay between base station and rover.

Although the RC and rover telemetry links both operate at 2.4GHz, they are independent systems. Simultaneously operating both links may potentially introduce signal interference, which we plan to mitigate by using channels at the opposite ends of the frequency band. However, this must be verified with additional experimentation.

This design ensures that the loss or degradation of one communication function would not impact others. More practical independence still needs to verify the system's requirements.

3.2.2 Link Budget Calculation Methodology

Link budgeting is a method of estimating the received signal power using transmitter output, antenna gains and signal attenuation. This provides an initial theoretical gauge of the range of the signal based on our selected hardware's specifications. In this analysis, unobstructed signal

attenuation is calculated using free-space path loss (FSPL), which increases with both transmission frequency and distance. We also account for additional losses through cables and other system imperfections. The resulting receiving power is compared against the receiver's estimated sensitivity to determine the link margin.

3.2.3 Link Budget Equations

Free Space Path Loss	$PL = 32.44 + 20lg(f) + 20lg(d)$ Where: • PL: Free space path loss (dB) • f: Frequency (MHz) • d: Distance (km)
RF Cable Loss	$L_{cable} = \alpha l$ Where: • α: Attenuation per unit length (dB/m) • l: Cable Length (m)
Received Signal Power	$P_{RX} = P_{TX} + G_{TX} + G_{RX} - PL - L$ Where: • P_{RX}: Received Power • P_{TX}: Transmitted Power • G_{RX}: Receiver Antenna Gain • G_{TX}: Transmitter Antenna Gain
Link Margin	$LM = P_{RX} - S_{RX}$ Where: • LM: Link Margin • S_{RX}: Receiver Sensitivity

3.2.4 Calculation Assumptions

Parameter	**Assumed value**	**Justification**
Operating distance	1.2 km per hop	A conservative value for the estimated 1km maximum distances between the base station, drone and lander.
Propagation model	Free-space path loss	Assuming unobstructed line-of-sight communication.
Minimum link margin	20 dB	Recommended conservative margin for Outdoor Long-Range RF Communications.

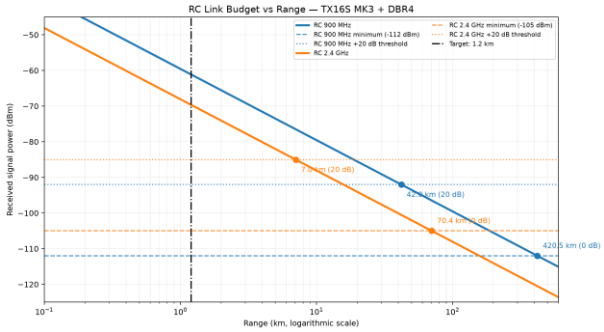
Antenna alignment	Nominal alignment	Assumes antennas operate with their stated gains in the relevant direction
Environmental conditions	Unobstructed LOS	Establishes theoretical performance before terrain and interference losses
Additional Losses	Link-Specific	Represents connector, cable and miscellaneous implementation losses.

3.2.5 RC Control Link

The RC link provides the longest calculated range among the three links considered. For this link we decided on the RadioMaster TX16S Mk3 Controller paired with the DBR4 Dual-band receiver. This system is advantageous for its dual-band capability, where both devices can switch between the primary 2.4GHz and the secondary 900MHz frequency bands. From the link budget table below, we find that a 900MHz link outperforms 2.4GHz in terms of range due to less free-space path loss.

Parameter	**2.4GHz**	**900MHz**	**Description**
Transmitter Power	30 dBm	33 dBm	Taken from Specs table
Transmitter Antenna Gain	2 dBi	2 dBi	“
Receiver Antenna Gain	2 dBi	2 dBi	“
Additional Losses	4.4 dB	2.6 dB	Cable Loss formula
Receiver Sensitivity	-105 dBm	-105 dBm	Preliminary estimation
Link Margin at 1.2km	**50.9 dB**	**35.4 dB**	**> 20 dB Minimum**

As shown in the table, both proposed links would meet the minimum link margin requirement. The consideration between using either 2.4GHz or 900MHz during the competition depends on local frequency band restrictions, actual range test performance and potential interference with other links from the base station and rover. This will be determined through additional testing conducted in semester 2.



To further illustrate the capabilities of the links, this graph shows the maximum range of the links while staying above the 20 dBm margin. As shown, the maximum range of the 2.4GHz and

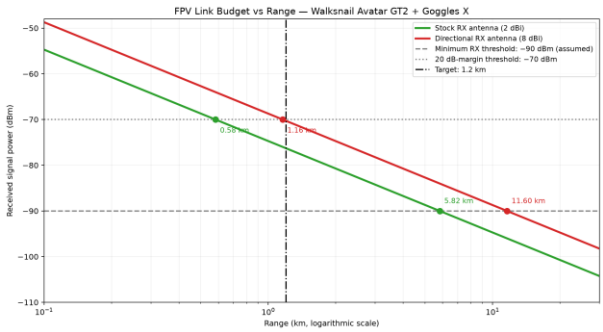
900MHz signals are 7.0km and 42.0km respectively. Hence, we conclude that this design meets the required range specifications.

3.2.6 FPV Video Link

Secondly, the FPV link ensures continuous transmission of live video feed throughout the missions, which is especially useful for reconnaissance. We chose the Walksnail GT2 Camera and (Video Transmitter) VTX kit, which outputs 1080p video at 60 frames-per-second. It is also compatible with the GM3 gimbal and our existing Walksnail Goggles X, reducing the need for additional system integration. However, the 5.8GHz link experiences the largest free-space attenuation at the same distance. Therefore, selecting a transmitter with sufficient output power and antenna gain is crucial to ensure reliable signal performance. The table below showcases the calculated link margins using the stock antenna and an 8 dBi antenna which increases the gain of the transmitter.

Parameter	**Stock Antenna**	**8 dBi Antenna**	**Description**
Transmitter Power	33 dBm	33 dBm	Taken from Specs table
Transmitter Antenna Gain	2 dBi	8 dBi	=
Receiver Antenna Gain	2 dBi	2 dBi	=
Additional Losses	2 dB	2 dB	Cable Loss formula
Receiver Sensitivity	-90 dBm	-90 dBm	Preliminary estimation
Link Margin at 1.2km	**13.7 dB**	**19.7 dB**	**< 20 dB Minimum**

As shown, theoretically an 8 dBi antenna would increase the link margin close to the minimum requirement at 1.2km. However, more experimental validation of this system performance is required before we confirm the purchase of this antenna.



This graph shows the theoretical range of the link with and without the upgraded antenna. From this we conclude that an upgraded antenna is required to reach the maximum drone distance for our missions.

3.2.7 Repeater Link

Lastly, our 2.4GHz repeater link provides the wifi extension required from the base station to the rover. We intend to use the GL-iNet GL-AR300M-Ext module for its compact form factor and the appropriate 2.4GHz wi-fi protocol to establish the connection. The link budget table below shows its specifications:

Parameter	**Value**	**Description**
Rocket M2 Transmitter Power	20 dBm	Assumed for mission requirements
Rocket M2 Transmitter Antenna Gain	12 dBi	Specs Sheet
GL-AR300M-Ext Transmitter Power	20 dBm	Assumed for mission requirements
GL-AR300M-Ext Receiver Antenna Gain	5 dBi	Specs Sheet
Additional Losses	2 dB	Cable Loss formula
Receiver Sensitivity	-90 dBm	Preliminary estimation
Link Margin at 1.2km	**23.4 dB**	**< 20 dB Minimum**

The relatively high margin of 23.4 dB suggests sufficient headroom is available for line-of-sight conditions. However, actual performance may be affected by cable attenuation, antenna orientation, interference and selected ELRS operating more. Hence performance validation is required.

3.2.8 Link Budget Summary

With the appropriate antennas, all three links theoretically have sufficient power to sustain the full distance for this competition. However, additional factors during actual flight such as antenna alignment, terrain obstruction and radio interference cannot be accounted for in calculations. As such, physical testing must be done to validate system performance.

3.3 Wiring Architecture

Figures 1 to 6 show the subsystem connections and their purposes. Orange arrows indicate power, blue indicates control or serial data, teal indicates Controller Area Network (CAN), and purple indicates radio or camera coax.

3.3.1 Overall System

3.3.2 Power Distribution System

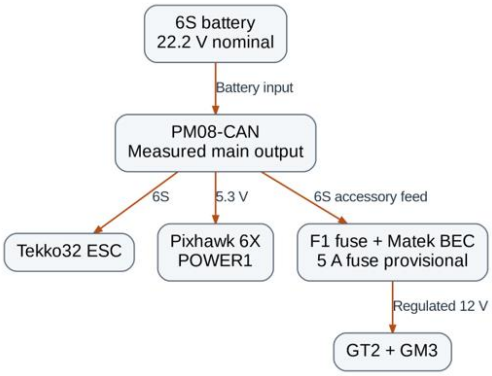


Figure 1 Battery power and regulated electronics supplies

The battery feeds the power module (PM08-CAN). Its main output supplies the ESC and fused accessory branch. The PM08-CAN also provides regulated 5.3 V to the flight controller, Pixhawk through an in-built power outlet and sends battery measurements separately through its CAN. The step-down converter (Matek BEC) is set to 12 V for the GT2 video transmitter and GM3 gimbal [5,6]. This prevents propulsion current from passing through the flight controller and avoids applying full battery voltage to the video equipment.

3.3.3 Propulsion System



Figure 2 Independent ESC channels and motor phase connections

The electronic speed controller (ESC), Tekko32 contains four independently controlled motor channels. Pixhawk sends a separate DShot300 command to each, and the ESC converts battery power into the three-phase supply required by each motor. The integrated board reduces separate mounts and power connections compared with four individual ESCs. Its tradeoffs are concentrated heat and whole-board replacement following a channel failure.

3.3.4 Flight Control System

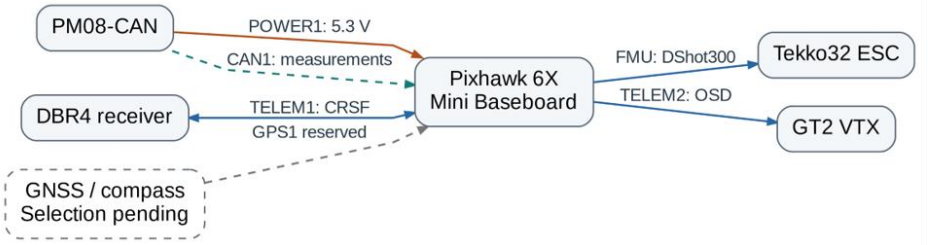


Figure 3 Pixhawk inputs and outputs with GNSS provision

Pixhawk receives pilot commands and sensor measurements, then ArduPilot calculates individual motor outputs to control attitude. The Pixhawk 6X provides serial, CAN and motor interfaces , the Mini Baseboard reduces mounting space and mass compared with building a custom baseboard [3]. The GT2 receives flight information separately for its on-screen display.

3.3.5 Radio Control System



Figure 4 Pilot commands and return telemetry through the DBR4

ExpressLRS, a communication protocol built for drones transmits commands to the DBR4, which forwards commands to Pixhawk. This command path is separate from rover relay.

3.3.6 Video and Gimbal System

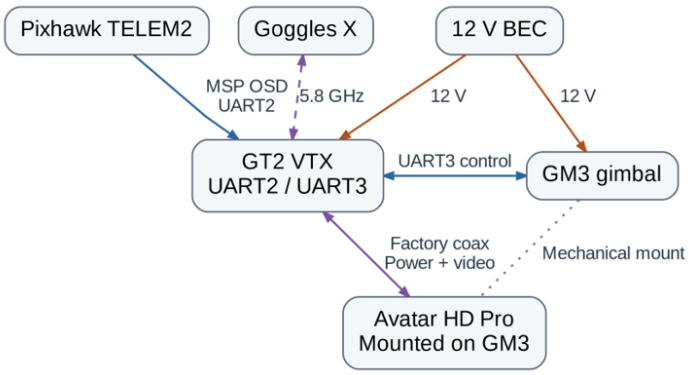


Figure 5 Camera video OSD and gimbal control paths

The camera exchanges power and image data with the GT2. The GT2 transmits video to the ground goggles, receives Pixhawk OSD information on UART2, and uses UART3 for gimbal control. The GM3 lets the operator redirect the camera while the drone maintains its relay position. This supports terrain reconnaissance and observation of the rover without turning the entire aircraft. The gimbal improves observation rather than directly increasing radio range [9,12].

3.4 Software Stack

3.5 Technical Challenges

3.5.1 Piloting from within the Base Station

One major design challenge was ensuring operation of the drone from within our base station. Rather than having a clear line-of-sight from the operator to the drone, the base station is an enclosed area, which cuts off radio connection via antenna. As such, we are required to find extension cables to place the antenna outside the base station.

The competition guidelines recommend a length of 20 meters of antenna cable from the base station to the antenna housing area. This poses some additional difficulties, as long distance cables introduce additional cable loss into our link budget calculations. For our design we selected a 20 meter LMR400 coaxial cable as the extension cable from the controller to its antenna and a similar length HDMI cable to transmit the video feed from the goggles to the monitor. Our calculated cable losses are as follows:

Cable	Attenuation Per Meter	Calculated Loss	Link Margin

LMR400 (at 2.4GHz)	0.218 dB/m	4.36 dB	32.8 dB
LMR400 (at 900MHz)	0.128 dB/m	2.56 dB	46.5 dB
HDMI (8 dBi antenna)	0.3 – 0.8 dB/m	6 – 16 dB	3.7 - 13.7 dB

The actual performance of both extension architecture remains uncertain due to several factors that are difficult to quantify through theoretical calculations. Firstly, the signal attenuation of the HDMI cable varies depending on the cable's construction and specifications. Furthermore, the use of LMR400 coaxial cables requires additional SMA to N-type adaptors at the controller and antenna connections. These also introduce additional insertion losses and potential impedance mismatches, which may further reduce the effectiveness of transmitted signal power.

While theoretical calculations provide an initial estimate of the attenuation, they may not fully account for the practical losses and its combined effects on the system performance. Consequently, these calculations alone are insufficient to establish the practical feasibility of the proposed architecture. Experimental validation is necessary, including RF range testing to verify stable video transmission over 20 meters, to determine suitability for actual deployment and if alternative cable solutions are required.

3.5.2 Estimated System Parameters

Our second major design challenge is having reliance on manufacturer datasheets for component specifications. Unlike other subsystems in our CCA, budget and time constraints limit our ability to develop and test multiple prototypes. Consequently, important parameters like component dimensions, mass, battery performance and motor efficiency are derived primarily from manufacturer data, where components are tested under controlled environments. These values may differ in actual operating conditions, which introduces uncertainty in our design.

This issue is further exacerbated by the interdependence of design parameters. One example is our battery mass, which determines the dummy payload's mass and affects flight endurance. Hence, an underestimation of battery mass would significantly impact our flight time estimation. Such dependencies mean that small inaccuracies would have cumulative impact on our drone's performance.

To mitigate uncertainty, we implemented conservative design margins into our calculation, as follows:

Parameter	**Value**
Overall Mass Growth	5%
Wiring Loss	10%
ESC Loss	10%
Battery Reserve	15%
Battery Voltage Sag Estimation	Dynamic value from internal resistance and current draw.
Minimum RF Link Margin	20 dB

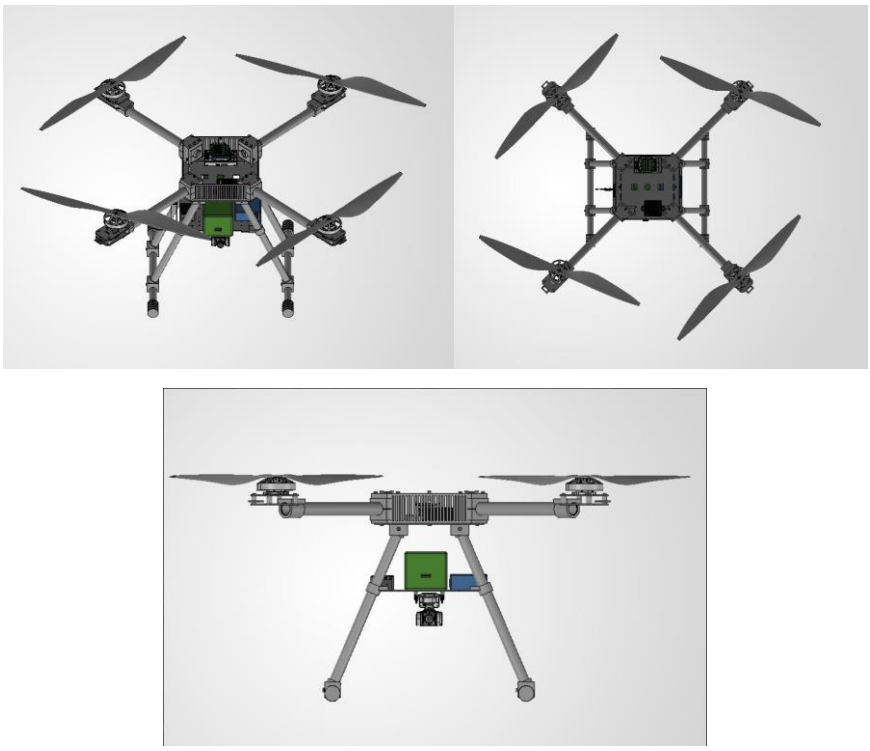
These assumptions aim to prevent overestimation of drone performance, providing conservative predictions of flight time and range. This is particularly importantly for mission planning, as it allows other teams to plan within realistic performance limits while reducing the risk of exceeding our actual capabilities. These estimates will be refined through further component validation and testing.

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4. Preliminary Testing

4.1 Mechanical Design and Integration

The symmetric four-arm frame is sized to accommodate 18-inch propellers. The central structure is divided into an upper electronics platform for the flight controller and communication hardware and a lower compartment for the battery and camera/gimbal. The battery is located near the geometric centre to reduce its contribution to centre-of-gravity offset; the full mass-weighted CG must still be calculated, including the equal dummy mass. The underslung camera supports downward terrain observation, while the landing gear provides ground clearance.



Preliminary CAD view of the component arrangement.

The initial CAD model was developed before the physical components were received. With the components now available, the next steps include 3D scanning components with complex geometries, refining mounting interfaces, verifying propeller clearance and wiring accessibility, determining the overall center of gravity (CG), and finalizing mechanical integration. Structural and flight-test validation have not yet been conducted at this interim stage.

4.2 Repeater Module Testing

Preliminary testing was conducted to evaluate compatibility between our selected GL-AR300-Ext Module and the Rocket M2 radio, which are part of the 2.4GHz communications relay between the base station, drone and lander. The objective was to establish a wireless connection between the two devices and verify basic network connectivity through ping testing.

Testing & Results

Testing between different configurations of the GL.iNet module, we successfully connected the Wi-Fi repeater in both router and extender modes, through reconfiguration of IP addresses. Our final ping test after rounds of troubleshooting successfully transmitted four packets at a response time of 2 to 30 milliseconds. These results demonstrate achievable connection between the two devices under short range conditions.

Further testing is required to ensure long-range communication with realistic outdoor operating conditions, and end-to-end relay functionality must also be validated before we can confirm the suitability of this architecture.

5. Project Timeline

We planned our project timeline around the various milestones in our project. With major milestones placed at the end of each semester, we broke down the work into 6 phases, ensuring our final milestone can be achieved within the academic year. The following table showcases the breakdown of each phase in our timeline and their desired outcomes.

S/N	Phase / Milestone	Semester	Description
1	Design	1	Defining of system requirements, communication hardware, establishing of mass and power budgets and initial CAD models.
2	Subsystem Testing	1	Testing of proposed repeater hardware with GNSS team, review of selected component's specifications, estimate of system performance.
	Interim Submission	**1**	**Review of Design & Initial Testing Phase**
3	Full Component Testing	Winter + 2	Finalise on CAD and FEA simulations, performance validation of all components.
4	Integration	2	Complete electrical and mechanical assembly and programming of flight controller and communications payload.
5	Flight Testing	2	PID Tuning, hover endurance testing, range testing, mission planning.
6	CV Integration	2	Benchmarking of different AI models, conducting integrated mission testing.
	Final Submission & Competition	**2**	**Review of design specifications and evaluation of system performance.**

5.1 Parallel Workstreams

For efficiency and clear definition of our personal responsibilities, we developed individual workstreams to concurrently work on different aspects of the design. We also set-up regularly scheduled team meetings and participated in biweekly CCA meetings to ensure component level integration and aligning of design perspectives with other subsystems. The following table describes our individual workstreams.

Phase	Design	Subsystem Testing	Component Testing	Integration	Flight Testing
Julian	RF Selection	Repeater Testing	FPV & Signal Testing	CV architecture Research & Software	CV & Range Testing

Joel	FC/ESC Selection	Wiring Architecture	FC/ESC Testing	Electrical Assembly & Software	PID Tuning
Benedict	Mass & Power Budget	Motor & Battery Sizing	Motor Thrust Testing	Mechanical Assembly	Endurance / Hover Testing
ZiHann		Frame Selection	CAD Designing	FEA analysis + Frame Build	Fine-tune component fits.

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6. Risk Analysis

Identified Hazard	Pre-Mitigation Risk	Mitigation Action	Post-Mitigation Risk
1. Mid-Flight Connection Loss	**High (Likely × Severe)**	Autonomous onboard failsafe scripts and redundant 900MHz telemetry backup link.	**Low (Unlikely × Minor)**
2. Premature Flight Crash	**Medium (Possible × Critical)**	3-stage progressive testing: No-prop motor test v Tethered rope hover v Free flight	**Low (Unlikely × Minor)**
3. Thermal Overheating	**High (Likely × Critical)**	External ESC prop-wash mounting, CNC heatsinks, RPM filtering, and 80°C telemetry threshold.	**Low (Unlikely × Moderate)**

Table (insert number): Summary of Identified Operational Hazards and Technical Mitigation Strategies.

To ensure safety and hardware protection during the building and testing phases of our drone, we evaluated potential failure modes and implemented targeted safeguards to eliminate or minimize their impact. Because early assembly and flight validation carry heightened risks of software misconfiguration, link instability, and thermal buildup, our development workflow prioritizes controlled, incremental verification and automated failsafe boundaries before advancing to unrestricted flight trials. Table (insert number) summarizes the primary hazards identified during our testing phase alongside their corresponding mitigation strategies and resulting risk reductions.

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7. Summary

To conclude, our project aims to develop a custom quadcopter for supporting the NUS Mars Rover Team in the University Rover Challenge 2027, with our main capabilities of aerial reconnaissance and establishing non-line-of-sight communications relay. Overall, we laid the foundation of our design in this semester through an iterative design process, and developed a clear plan for full integration and operational testing of the drone in semester 2.

Current Progress

Throughout this first semester, our team established our preliminary system architecture and component selections, conducting subsystem-level analysis to evaluate the feasibility of the proposed design. Significant progress was made in propulsion and battery optimisation, mechanical design, electrical system integration and wireless communication architecture. Using iterative mass budgeting, flight endurance estimation, and link budgeting calculations, we developed a design that both achieves mission objectives and stays within competition restrictions. Initial wireless repeater connectivity has also been successfully established in this term.

Future Work

Nevertheless, uncertainties remain in terms of actual flight performance, due to a heavy reliance on manufacturer specifications and theoretical calculation frameworks. While conservative design margins have been incorporated to account for our potential discrepancies, further experimental validation required to verify actual performance and reliability. In semester 2, we aim to complete mechanical and electrical integration of the parts, along with individual component testing, before proceeding with flight tests, CV integration and ultimately mission planning for the final competition. These activities will help to address our uncertainties in flight performance and evaluate the system against our initial design requirements, to ensure an operationally ready drone before our final submission and URC2027 Competition.

References

URC2027 Guidelines: https://urc.marssociety.org/home/requirements-guidelines